Binomial Expansion

Starters

Solution 2

1. Expand to the term in x^3.

(a) $(1+5x)^7$

1a $1 + 35x + 525x^2 + 4375x^3 + \cdots$

(b) $(5-2x)^{10}$

1b
$9765625 - 39062500x$
$+ 70312500x^2 - 75000000x^3 + \cdots$

(c) $(1-x)(2+x)^6$

1c $64 + 128x + 48x^2 - 80x^3 + \cdots$

2. Express in partial fractions.

(a) $\dfrac{-14x+7}{(1+2x)(1-5x)}$

2a
$$\frac{4}{1+2x} + \frac{3}{1-5x}$$

(b) $\dfrac{24x-1}{(1+2x)^2}$

2b
$$\frac{12}{1+2x} - \frac{13}{(1+2x)^2}$$

(c) $\dfrac{24x^2+48x+24}{(1+x)(3x-4)^2}$

2c
Cancel $1+x$, then split:
$$\frac{8}{3x-4} + \frac{56}{(3x-4)^2}.$$

Chapter Overview

Solution 3

1. New powers

We extend the binomial expansion to negative and fractional powers.

$\sqrt{1+x} = (1+x)^{1/2}$.

The expansion is generally an infinite series, with a restricted interval of validity.

2. A constant other than 1

First factor out the constant:

$$(8+5x)^{-1/3} = \frac{1}{2}\left(1+\frac{5x}{8}\right)^{-1/3}.$$

Then use the binomial formula.

3. Rational expressions

Partial fractions can give simpler expressions to expand:

$$\frac{4-5x}{(1+x)(2-x)}.$$

Expand each fraction, then combine like powers.

Before You Start

Solution 4

Before you start

For every expansion:

1. Identify the power and the expression replacing x.
2. Keep brackets around negative terms.
3. Simplify one coefficient at a time.
4. State the interval of validity.
5. For an approximation, check that the substituted value lies in that interval.

Pure Year 1 Recap

Solution 5

Year 1 recap

Expand each expression.

$(1+x)^5$

[1] $1+5x+10x^2+10x^3+5x^4+x^5$

$(1+2x)^4$

$1+8x+24x^2+32x^3+16x^4$

$(1-3x)^3$

$1-9x+27x^2-27x^3$

Generalised coefficients

$$\binom{n}{r}=\frac{n(n-1)\cdots(n-r+1)}{r!}.$$

This formula also makes sense for non-integer n.

Evaluate $\binom{10}{3}$, $\binom{-1}{2}$, $\binom{-2}{3}$, $\binom{1/2}{2}$.

$$\binom{10}{3}=120,\quad \binom{-1}{2}=1,$$
$$\binom{-2}{3}=-4,\quad \binom{1/2}{2}=-\frac{1}{8}.$$

Binomial Expansion

Solution 6

The binomial series

$$(1+x)^n=1+nx+\frac{n(n-1)}{2!}x^2+\frac{n(n-1)(n-2)}{3!}x^3+\cdots.$$

For negative or fractional n, use $|x|<1$. For non-negative integer n, the expansion terminates and is valid for every real x.

Example 1

Expand $(1+x)^{-1}$ to x^3.

[1] $n=-1$:

$1-x+x^2-x^3+\cdots,\quad |x|<1.$

Example 2

Expand $\sqrt{1-3x}$ to x^3.

Use $n=\frac{1}{2}$ and replace x by $-3x$:

$1-\frac{3}{2}x-\frac{9}{8}x^2-\frac{27}{16}x^3+\cdots.$

Valid for $|x|<\frac{1}{3}$.

When are infinite expansions valid? Solution 7

Why does validity matter?

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots.$$

The right side is a geometric series with ratio $-x$.

For which values of x does it converge?

Validity

A geometric series converges when its ratio has magnitude less than 1:

$$|-x| < 1 \iff -1 < x < 1.$$

Compare these cases

$0 < x < 1$: signs alternate and magnitudes decrease.

$-1 < x < 0$: all terms are positive and decrease.

$x = 1$: partial sums alternate between 1 and 0.

$x > 1$: term magnitudes grow.

$x = -1$: the original fraction is undefined.

Decreasing terms alone do not prove convergence; here the geometric-series result does.

When are infinite expansions valid? Solution 8

Replace the whole expression

For $\sqrt{1-3x}$, require

$$|-3x| < 1 \iff |x| < \frac{1}{3}.$$

State the standard interval of validity for each expansion.

(a) $(1+2x)^{-1}$

(b) $(1-x)^{-2}$

(c) $(1+x/4)^{1/2}$

(d) $(1-2x/3)^{-1}$

a (a) $|2x| < 1 \Rightarrow |x| < \frac{1}{2}$.

b (b) $|-x| < 1 \Rightarrow |x| < 1$.

c (c) $|x/4| < 1 \Rightarrow |x| < 4$.

d

(d) $|2x/3| < 1 \Rightarrow |x| < \frac{3}{2}$.

Combining Expansions

Solution 9

Edexcel C4 June 2013, Q2

Show that, for small x,

$$\sqrt{\frac{1+x}{1-x}} \approx 1 + x + \frac{1}{2}x^2.$$

Write the expression as $(1+x)^{1/2}(1-x)^{-1/2}$.

Expand

$$(1+x)^{1/2} = 1 + \frac{x}{2} - \frac{x^2}{8} + \cdots,$$

$$(1-x)^{-1/2} = 1 + \frac{x}{2} + \frac{3x^2}{8} + \cdots.$$

Combine

Multiply, keeping powers up to x^2:

$$1 + x + \left(-\frac{1}{8} + \frac{1}{4} + \frac{3}{8}\right)x^2.$$

$$\boxed{1 + x + \frac{1}{2}x^2}$$

Both series are valid for $|x| < 1$.

Test Your Understanding

Solution 10

1. Expand $(1+4x)^{-2}$ to x^3, and state its interval of validity.

1

$$1 - 8x + 48x^2 - 256x^3 + \cdots.$$

Valid for $|4x| < 1$, so $|x| < \frac{1}{4}$.

2. Edexcel C4 January 2010, Q1

(a) Expand $\sqrt{1-8x}$ to x^3.

(b) Show that at $x = 1/100$, the exact value is $\sqrt{23}/5$.

(c) Hence approximate $\sqrt{23}$ to 5 decimal places.

a

(a) $1 - 4x - 8x^2 - 32x^3 + \cdots$.

(b) $\sqrt{1 - 8/100} = \sqrt{23/25} = \sqrt{23}/5$.

(c) At $x = 0.01$, the series gives 0.959168:

$\sqrt{23} \approx 5(0.959168) = 4.79584$.

Accuracy of an approximation

Solution 11

Accuracy of a truncated series

Approximate $1/(1+x)$ using $1 - x + x^2$, with $x = 0.01$.

Approximation

$1 - 0.01 + 0.0001 = 0.9901.$

The exact value is $1/1.01 \approx 0.9900990099.$

Absolute error:

$$\left|0.9901 - \frac{1}{1.01}\right| \approx 9.901 \times 10^{-7}.$$

Why is this close?

The first omitted term is $-x^3 = -10^{-6}$, and the later terms are smaller.

Using more terms usually improves an approximation within the interval of convergence.

Values near the boundary can require many more terms for comparable accuracy.

Exercise 4A pages 96-97

Solution 12

Exercise 4A, Question 4

$f(x) = \dfrac{1+x}{1-2x}$.

(a) Find the expansion in ascending powers of x, up to and including x^3.

(b) State the interval of validity.

a

$$\frac{1}{1-2x} = 1 + 2x + 4x^2 + 8x^3 + \cdots.$$

Multiply by $1 + x$:

$$\boxed{f(x) = 1 + 3x + 6x^2 + 12x^3 + \cdots}.$$

b

$|2x| < 1$, hence $\boxed{|x| < \frac{1}{2}}$.

Exercise 4A, Question 5

$f(x) = \sqrt{1+3x}, \qquad -\frac{1}{3} < x < \frac{1}{3}.$

(a) Expand to x^3.

(b) Show that $f(0.01) = \sqrt{103}/10$.

(c) Find the percentage error in the series approximation at $x = 0.01$, to 2 significant figures.

a

$$f(x) = 1 + \frac{3}{2}x - \frac{9}{8}x^2 + \frac{27}{16}x^3 + \cdots.$$

b

$$f(0.01) = \sqrt{1.03} = \frac{\sqrt{103}}{10}.$$

c

The approximation is $p = 1.0148891875$.

The exact value is $v = \sqrt{1.03}$.

$$\text{Percentage error} = \frac{|p-v|}{v} \times 100\%.$$

This is approximately 0.0000031%.

Exercise 4A, Question 6

In $(1+ax)^{-1/2}$, the coefficient of x^2 is 24.

(a) Find the possible values of a.

(b) Find the corresponding coefficient of x^3.

6

$$\frac{(-1/2)(-3/2)}{2}a^2 = \frac{3}{8}a^2 = 24.$$

Hence $a^2 = 64$, so $a = \pm 8$.

The cubic coefficient is $-5a^3/16$:

$a = 8 \Rightarrow -160$; $a = -8 \Rightarrow 160$.

Question 7

Show that, for small x,

$$\sqrt{\frac{1+x}{1-x}} \approx 1 + x + \frac{1}{2}x^2.$$

7

$$(1+x)^{1/2} = 1 + \frac{x}{2} - \frac{x^2}{8} + \cdots,$$

$$(1-x)^{-1/2} = 1 + \frac{x}{2} + \frac{3x^2}{8} + \cdots.$$

Multiply and collect terms:

$$1 + x + \left(-\frac{1}{8} + \frac{1}{4} + \frac{3}{8}\right)x^2 = 1 + x + \frac{1}{2}x^2.$$

Exercise 4A pages 96-97

Solution 15

Exercise 4A, Question 8

$h(x) = \dfrac{6}{1+5x} - \dfrac{4}{1-3x}$.

(a) Expand to x^2. (6 marks)

(b) Find the percentage error when the expansion estimates $h(0.01)$, to 2 significant figures. (3 marks)

(c) Explain why it cannot be used for $h(0.5)$. (1 mark)

a

$$h(x) = 6(1 - 5x + 25x^2 + \cdots) - 4(1 + 3x + 9x^2 + \cdots) = 2 - 42x + 114x^2 + \cdots.$$

b

Approximation: 1.5914. Exact: $6/1.05 - 4/0.97$.

Percentage error $\approx 0.052\%$.

c

Both expansions require $|x| < \frac{1}{5}$.

0.5 is outside this interval.

Exercise 4A pages 96-97

Solution 16

Exercise 4A, Question 9

(a) Expand $(1-3x)^{3/2}$ to x^3. (4 marks)

(b) Show that at $x = 1/100$, its exact value is $97\sqrt{97}/1000$. (2 marks)

(c) Hence approximate $\sqrt{97}$ to 5 decimal places. (3 marks)

a

$$(1-3x)^{3/2} = 1 - \frac{9}{2}x + \frac{27}{8}x^2 + \frac{27}{16}x^3 + \cdots.$$

b

$$\left(\frac{97}{100}\right)^{3/2} = \frac{97\sqrt{97}}{1000}.$$

c

At $x = 0.01$, the series gives 0.9553391875.

$$\sqrt{97} \approx \frac{1000(0.9553391875)}{97} = 9.84886.$$

Factoring Out the Constant

Solution 17

When the constant is not 1

Expand $\sqrt{4+x}$ to x^3, and state the interval of validity.

First factor out 4 inside the root:

$$\sqrt{4+x} = 2\left(1+\frac{x}{4}\right)^{1/2}.$$

Expand

$$2\left[1+\frac{1}{2}\frac{x}{4}-\frac{1}{8}\left(\frac{x}{4}\right)^2+\frac{1}{16}\left(\frac{x}{4}\right)^3+\cdots\right].$$

$$\boxed{2+\frac{x}{4}-\frac{x^2}{64}+\frac{x^3}{512}+\cdots}.$$

Validity

$$\left|\tfrac{x}{4}\right|<1 \Rightarrow \boxed{|x|<4}.$$

Quickfire First Step

Solution 18

Rewrite first, then state validity.

(a) $(2+x)^{-3}$

a
$\frac{1}{8}(1+x/2)^{-3}$, $|x|<2$.

(b) $(9+2x)^{1/2}$

b $3(1+2x/9)^{1/2}$, $|x|<9/2$.

(c) $(8-x)^{1/3}$

c
$2(1-x/8)^{1/3}$, $|x|<8$.

(d) $(5-2x)^{-3}$

d $\frac{1}{125}(1-2x/5)^{-3}$, $|x|<5/2$.

(e) $(16+3x)^{-1/2}$

e $\frac{1}{4}(1+3x/16)^{-1/2}$, $|x|<16/3$.

Check the prefactor

Apply the power to the factored constant as well: $16^{-1/2}=1/4$.

Common Errors

Solution 19

Avoid these common errors

Signs

Square the whole term:

$(-3x)^2 = 9x^2, \qquad (-3x)^3 = -27x^3.$

Factorials

The cubic coefficient uses $3! = 6$, not 3.

Brackets

In $(1 - 3x)^n$, every power applies to $-3x$.

Constant factor

In $(a + bx)^n$, first write

$$a^n \left(1 + \frac{bx}{a}\right)^n.$$

Validity

Test $|bx/a| < 1$, not just $|x| < 1$.

Test Your Understanding

Solution 20

Edexcel C4 June 2013 (withdrawn), Q1

(a) Expand $\sqrt{9 + 8x}$, $|x| < 9/8$, up to and including x^2. Give simplified fractional coefficients. (5 marks)

(b) Use the expansion to estimate $\sqrt{11}$ as a single fraction. (3 marks)

a

$$\sqrt{9 + 8x} = 3\left(1 + \frac{8x}{9}\right)^{1/2}.$$

$$3\left[1 + \frac{4x}{9} - \frac{1}{8}\left(\frac{8x}{9}\right)^2 + \cdots\right].$$

$$\boxed{3 + \frac{4}{3}x - \frac{8}{27}x^2 + \cdots}.$$

b

Choose $x = 1/4$, so $9 + 8x = 11$:

$$\sqrt{11} \approx 3 + \frac{1}{3} - \frac{1}{54} = \boxed{\frac{179}{54}}.$$

Exercise 4B, Pages 99-100 — Solution 21

Exercise 4B, Question 2

$f(x) = (5+4x)^{-2}, \qquad |x| < \frac{5}{4}.$

Expand in ascending powers of x, up to and including x^3. Give simplified fractional coefficients. (5 marks)

Answer

$$f(x) = \frac{1}{25}\left(1+\frac{4x}{5}\right)^{-2}.$$

Use $(1+u)^{-2} = 1 - 2u + 3u^2 - 4u^3 + \cdots$:

$$\frac{1}{25}\left[1 - \frac{8x}{5} + \frac{48x^2}{25} - \frac{256x^3}{125} + \cdots\right].$$

$$\boxed{\frac{1}{25} - \frac{8}{125}x + \frac{48}{625}x^2 - \frac{256}{3125}x^3 + \cdots}.$$

Exercise 4B, Pages 99-100 — Solution 22

Exercise 4B, Question 3

$m(x) = \sqrt{4-x}, \qquad |x| < 4.$

(a) Expand to x^2. (4 marks)

(b) Show that $m(1/9) = \sqrt{35}/3$. (2 marks)

(c) Use (a) to approximate $\sqrt{35}$, and calculate the percentage error. (4 marks)

a

$$m(x) = 2(1-x/4)^{1/2} = 2 - \frac{x}{4} - \frac{x^2}{64} + \cdots.$$

b

$$m(1/9) = \sqrt{4-1/9} = \sqrt{35/9} = \frac{\sqrt{35}}{3}.$$

c

$$\sqrt{35} \approx 3\left(2 - \frac{1}{36} - \frac{1}{5184}\right)$$
$$= \frac{10223}{1728} \approx 5.916087963.$$

With $p = 10223/1728$, the percentage error is

$$\frac{|p-\sqrt{35}|}{\sqrt{35}} \times 100\% \approx 0.0001383\%.$$

Exercise 4B, Pages 99-100 — Solution 23

Exercise 4B, Question 4

The first three terms are

$$\frac{1}{\sqrt{a+bx}} = 3 + \frac{1}{3}x + \frac{1}{18}x^2 + \cdots .$$

(a) Find a and b.

(b) Find the coefficient of x^3.

a

$$\frac{1}{\sqrt{a+bx}} = a^{-1/2}\left(1 + \frac{b}{a}x\right)^{-1/2}.$$

Constant term: $a^{-1/2} = 3 \Rightarrow a = 1/9$.

Linear coefficient: $-b/(2a^{3/2}) = 1/3$.

$$\boxed{b = -\frac{2}{81}}.$$

b

The expansion is $3(1 - 2x/9)^{-1/2}$.

Its cubic coefficient is

$$3\left(-\frac{5}{16}\right)\left(-\frac{2}{9}\right)^3 = \boxed{\frac{5}{486}}.$$

Exercise 4B, Pages 99-100 — Solution 24

Exercise 4B, Question 5

$$f(x) = \frac{3 + 2x - x^2}{4 - x}.$$

Prove that, for sufficiently small x,

$$f(x) \approx \frac{3}{4} + \frac{11}{16}x - \frac{5}{64}x^2.$$

Answer

$$\frac{1}{4-x} = \frac{1}{4}\left(1 - \frac{x}{4}\right)^{-1} = \frac{1}{4} + \frac{x}{16} + \frac{x^2}{64} + \cdots .$$

Multiply by $3 + 2x - x^2$.

Constant: $3/4$.

Linear: $3/16 + 2/4 = 11/16$.

Quadratic: $3/64 + 2/16 - 1/4 = -5/64$.

$$\boxed{f(x) \approx \frac{3}{4} + \frac{11}{16}x - \frac{5}{64}x^2}.$$

The series is valid for $|x| < 4$.

Exercise 4B

Solution 25

AEA 2006, Question 1

(a) Expand $(1-y)^{-2}$ to y^3.

(b) Express the infinite series as $(a+x)^n$:

$$1+\frac{2x}{1+x}+\frac{3x^2}{(1+x)^2}+\cdots+\frac{rx^{r-1}}{(1+x)^{r-1}}+\cdots.$$

(c) State the values of x for which this is valid.

a

$$(1-y)^{-2}=1+2y+3y^2+4y^3+\cdots.$$

b

Set $y=x/(1+x)$. Then

$$\left(1-\frac{x}{1+x}\right)^{-2}=\left(\frac{1}{1+x}\right)^{-2}=(1+x)^2.$$

Thus $a=1$, $n=2$.

c

$$\left|\frac{x}{1+x}\right|<1 \iff x^2<(1+x)^2 \iff x>-\frac{1}{2}.$$

Using Partial Fractions

Solution 26

Expand using partial fractions

Expand to x^3 and state validity:

$$\frac{4-5x}{(1+x)(2-x)}.$$

Partial fractions

$$\frac{4-5x}{(1+x)(2-x)}=\frac{3}{1+x}-\frac{2}{2-x}.$$

Equating numerators:

$$4-5x=3(2-x)-2(1+x).$$

Expansion

$$3(1-x+x^2-x^3+\cdots)-\left(1+\frac{x}{2}+\frac{x^2}{4}+\frac{x^3}{8}+\cdots\right)$$
$$=2-\frac{7}{2}x+\frac{11}{4}x^2-\frac{25}{8}x^3+\cdots.$$

Validity

The first series requires $|x|<1$, the second $|x|<2$.

Both hold when $\boxed{|x|<1}$.

Test Your Understanding

Solution 27

Edexcel C4 June 2010, Q5

$$\frac{2x^2+5x-10}{(x-1)(x+2)} \equiv A + \frac{B}{x-1} + \frac{C}{x+2}.$$

(a) Find A, B, C. (4 marks)

(b) Expand to x^2, with simplified fractional coefficients. (7 marks)

a

$$2x^2+5x-10 = A(x-1)(x+2) + B(x+2) + C(x-1).$$

Compare x^2: $A = 2$.

Set $x = 1$: $-3 = 3B \Rightarrow B = -1$.

Set $x = -2$: $-12 = -3C \Rightarrow C = 4$.

b

$$2 - \frac{1}{x-1} + \frac{4}{x+2} = 2 + (1-x)^{-1} + 2(1+x/2)^{-1}$$

$$= 5 + \frac{3}{2}x^2 + \cdots.$$

Valid for $|x| < 1$.

Exercise 4C Pages 102-103

Solution 28

Exercise 4C, Question 1

(a) Express in partial fractions:
$\dfrac{8x+4}{(1-x)(2+x)}$.

(b) Expand to x^2.

(c) State the interval of validity.

a

$$\frac{8x+4}{(1-x)(2+x)} = \frac{4}{1-x} - \frac{4}{2+x}.$$

Check: $4(2+x) - 4(1-x) = 8x+4$.

b

$$4(1+x+x^2+\cdots) - 2(1-x/2+x^2/4+\cdots)$$

$$= \boxed{2 + 5x + \frac{7}{2}x^2 + \cdots}.$$

c

$|x| < 1$ and $|x| < 2$, so $\boxed{-1 < x < 1}$.

Exercise 4C Pages 102-103 — Solution 29

Exercise 4C, Question 2

(a) Express $-2x/(2+x)^2$ in partial fractions.

(b) Determine B, C in its expansion

$-\frac{1}{2}x + Bx^2 + Cx^3 + \cdots$.

(c) State validity.

a

$$-\frac{2x}{(2+x)^2} = -\frac{2}{2+x} + \frac{4}{(2+x)^2}.$$

b

$$\begin{aligned} &-(1+x/2)^{-1} + (1+x/2)^{-2} \\ &= -\left(1 - \frac{x}{2} + \frac{x^2}{4} - \frac{x^3}{8}\right) \\ &\quad + 1 - x + \frac{3x^2}{4} - \frac{x^3}{2} + \cdots \\ &= -\frac{x}{2} + \frac{x^2}{2} - \frac{3x^3}{8} + \cdots. \end{aligned}$$

$$\boxed{B = 1/2,\ C = -3/8}.$$

c

$$\boxed{|x| < 2}.$$

Exercise 4C Pages 102-103 — Solution 30

Exercise 4C, Question 4

$$g(x) = \frac{12x-1}{(1+2x)(1-3x)}, \qquad |x| < \frac{1}{3}.$$

(a) Find A, B such that

$$g(x) = \frac{A}{1+2x} + \frac{B}{1-3x}.$$

(b) Expand $g(x)$ to x^2.

a

$12x - 1 = A(1-3x) + B(1+2x)$.

Therefore $A + B = -1$, $-3A + 2B = 12$.

$$\boxed{A = -\frac{14}{5}, \quad B = \frac{9}{5}}.$$

b

$$\begin{aligned} g(x) &= -\frac{14}{5}(1 - 2x + 4x^2 + \cdots) \\ &\quad + \frac{9}{5}(1 + 3x + 9x^2 + \cdots) \\ &= \boxed{-1 + 11x + 5x^2 + \cdots}. \end{aligned}$$

Exercise 4C, Question 6

$$\frac{3x^2+4x-5}{(x+3)(x-2)} \equiv A + \frac{B}{x+3} + \frac{C}{x-2}.$$

(a) Find A, B, C. (4 marks)

(b) Expand to x^2, giving simplified fractional coefficients. (7 marks)

a

Compare quadratic coefficients: $A = 3$.

At $x = -3$: $10 = -5B \Rightarrow B = -2$.

At $x = 2$: $15 = 5C \Rightarrow C = 3$.

b

$$3 - \frac{2}{3}(1 + x/3)^{-1} - \frac{3}{2}(1 - x/2)^{-1}$$

$$= 3 - \frac{2}{3}\left(1 - \frac{x}{3} + \frac{x^2}{9}\right)$$

$$- \frac{3}{2}\left(1 + \frac{x}{2} + \frac{x^2}{4}\right) + \cdots$$

$$= \boxed{\frac{5}{6} - \frac{19}{36}x - \frac{97}{216}x^2 + \cdots}.$$

Valid for $|x| < 2$.